

PROGRAMMING ARITHMETIC AND LOGIC OPERATIONS

Implementation of Arithmetic and Logic Operations

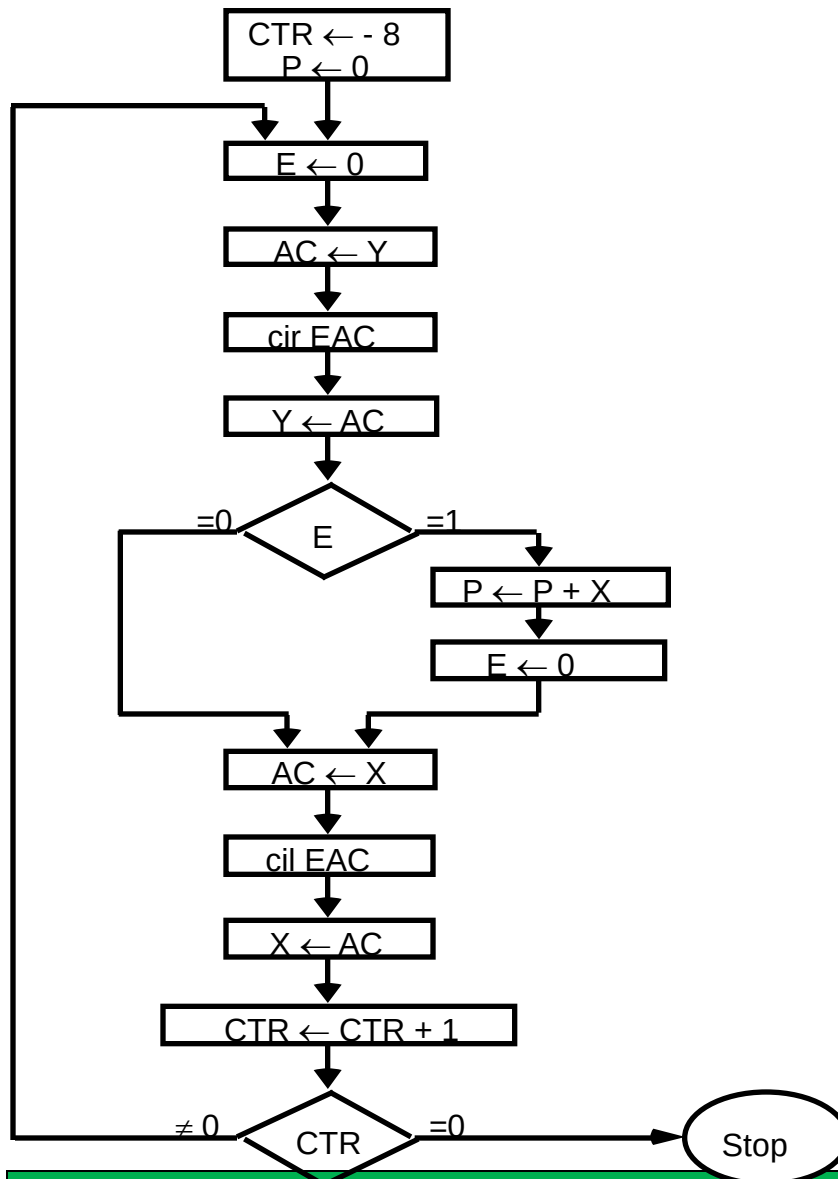
- Software Implementation
 - Implementation of an operation with a program using machine instruction set
 - Usually when the operation is not included in the instruction set
- Hardware Implementation
 - Implementation of an operation in a computer with one machine instruction

Software Implementation example:

- * Multiplication
 - For simplicity, unsigned positive numbers
 - 8-bit numbers -> 16-bit product



FLOWCHART OF A PROGRAM - Multiplication -



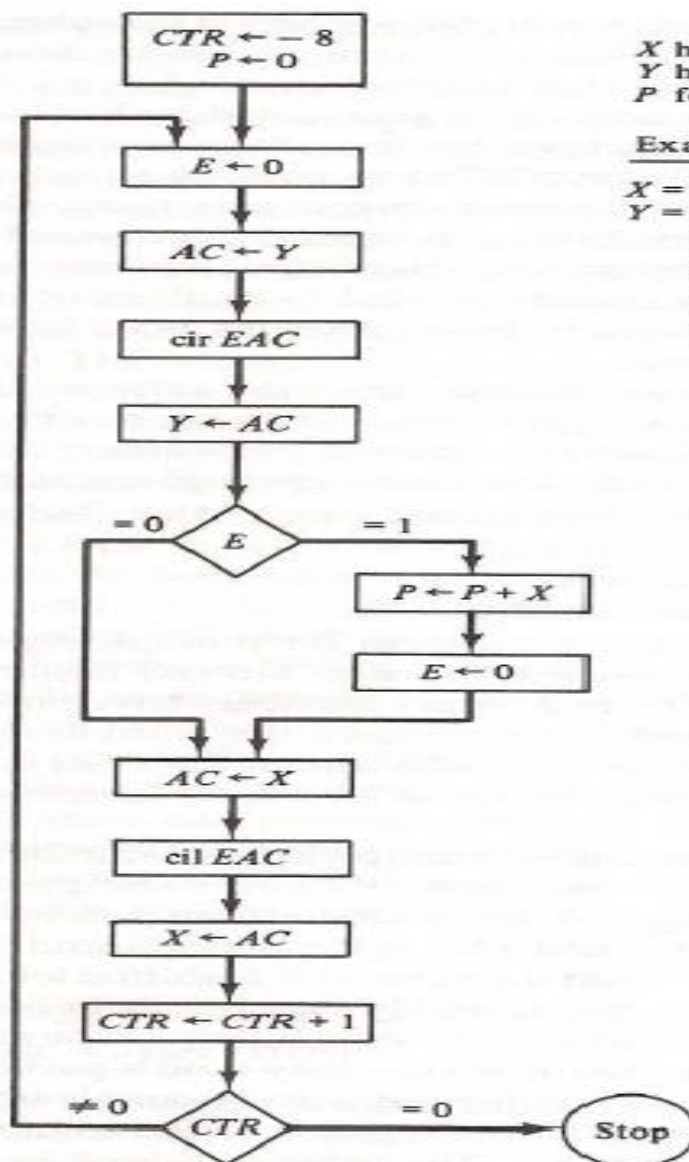
X holds the multiplicand
Y holds the multiplier
P holds the product

Example with four significant digits

X = 0000 1111	P
Y = <u>0000 1011</u>	0000 0000
0000 1111	0000 1111
0001 1110	0010 1101
0000 0000	0010 1101
<u>0111 1000</u>	1010 0101
1010 0101	



FLOWCHART OF A PROGRAM - Multiplication -



X holds the multiplicand
Y holds the multiplier
P forms the product

Example with four significant digits

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1010 0101	



ASSEMBLY LANGUAGE PROGRAM - Multiplication -

TABLE 6-14 Program to Multiply Two Positive Numbers

	ORG 100	
LOP,	CLE	/Clear E
	LDA Y	/Load multiplier
	CIR	/Transfer multiplier bit to E
	STA Y	/Store shifted multiplier
	SZE	/Check if bit is zero
	BUN ONE	/Bit is one; go to ONE
	BUN ZRO	/Bit is zero; go to ZRO
ONE,	LDA X	/Load multiplicand
	ADD P	/Add to partial product
	STA P	/Store partial product
	CLE	/Clear E
ZRO,	LDA X	/Load multiplicand
	CIL	/Shift left
	STA X	/Store shifted multiplicand
	ISZ CTR	/Increment counter
	BUN LOP	/Counter not zero; repeat loop
	HLT	/Counter is zero; halt
CTR,	DEC -8	/This location serves as a counter
X,	HEX 000F	/Multiplicand stored here
Y,	HEX 000B	/Multiplier stored here
P,	HEX 0	/Product formed here
	END	

